

UBC Course Data Analysis and Future Planning



Hackathon Team 11: Thunder Hackers

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February 21, 2025



THE UNIVERSITY OF BRITISH COLUMBIA





**Have you ever struggled with
course enrollment ?**

Business Question:

**How can we use data to assess and improve
UBC's course planning?**



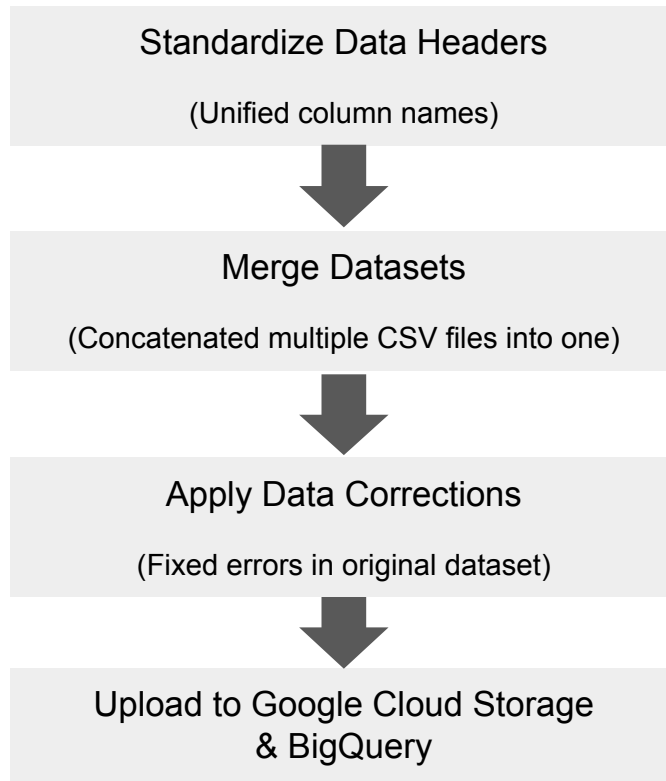
Data Overview

Data Sources: Github Public Data^[1] (10,000 Files, 350,000 rows)

- Covers UBC Vancouver Campus (1996S-2023W)
- **20 Key Features Extracted:**
 - Course Details (Faculty, Year & Session, etc)
 - Course Enrollments
 - Student Grade Distributions (A+, A, A-, B+, etc)
 - Instructor assignments

Final Dataset: ubc_courses_overall.csv

Data Cleaning & Peprocessing:



UBC Academics Dashboard

Enrollment Count

407,725

Tools Used:

BigQuery

For advanced
SQL-based filtering
& data transformation

Looker Studio

Interactive
dashboard creation

Google Vertex AI

Sentiment analysis on
course reviews

ARIMA Model

Future course
enrollment
predictions

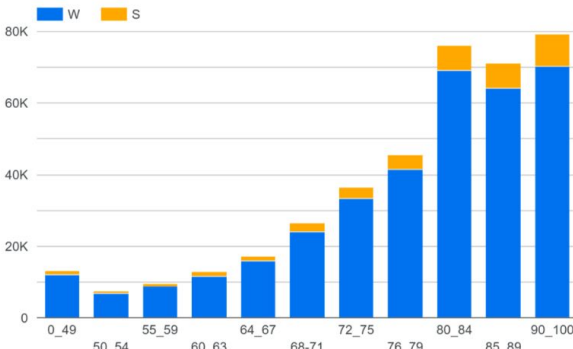
Course Performance Metrics Rank

Course Title	Fail Rate (%) ▾	High Score Rate (%)
1. Introduction to Rings and Modules	38.46	0
2. Early Modern Philosophy	36.84	10.53
3. Syntactic Theory and Analysis	33.33	0
4. Engineering Analysis I	28.57	21.43
5. Advanced Topics in Comparative Politics: No...	27.27	18.18
6. Integral Calculus with Applications	26.32	4.62
7. Topics in Number Theory	26.32	0
8. Neuromuscular Integration of Human Movem...	25.26	11.58
9. The British Empire after 1850	23.53	8.82
10. Matter and Energy I	21.43	7.14
11. Modern Persian Literature	20.59	14.71
12. Italian Fascism in Interdisciplinary Perspectiv...	20	20
13. Evidence-Based Genomic Counselling	20	53.33
14. Italian Literature and Culture of the Medieval ...	20	40
15. Foundations of Real Estate Mathematics	19.5	3.76
16. Thermal Physics I	19.27	14.68
17. Post-Revolutionary Iranian Cinema	18.52	3.7
18. Advanced Organic Chemistry	18.37	0
19. Philosophy of Biology	18.18	9.09
20. Themes and Interpretive Issues in Modern H...	18.18	18.18
21. British Columbia Real Estate Law and Real E...	17.81	2.51
22. Europe in the Age of the Enlightenment	17.65	0
23. Sensorimotor Control of Human Movement	17.24	17.24
24. Systematic Program Design	16.94	11.29
25. Phase Transformations	16.92	4.62

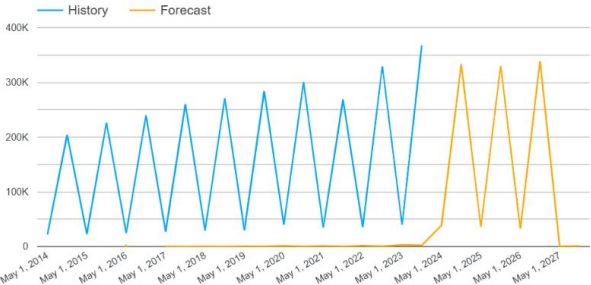
Top 3 Related Courses by Enrollment

	Correlation ▾
1. Capital Markets and Real Estate	0.96
2. British Columbia Real Estate Law and Real Estate Ethics	0.76
3. Business Economics	0.7

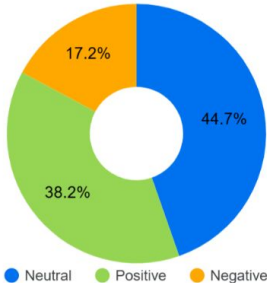
Student Score Distribution



Enrollment over Time with Forecasting



Course Reviews Sentiment



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Our Product

Analysis Workflow

Correlation Analysis

How enrollment & performance of one course would affect others?

Online Impact

Which courses work best in an online setting

Course Enrollment Trends

What are most / least popular courses?

Student Performance Analysis

Which courses have the highest fail rate?



Enrollment Trends: Most & Least Popular Courses

16x

Increase since 2014

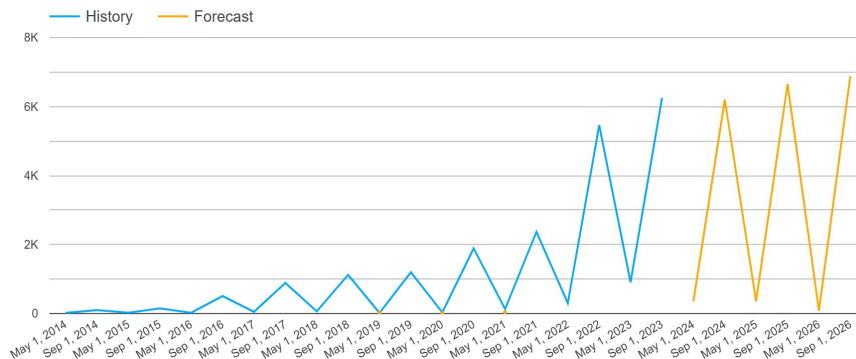
Principles of Forest Science
Scholarly Writing and Argumentation in Forestry
Stress and Strategies to Promote Well Being
Application of Statistics in Business
Statistical Inference for Data Science
Data Visualization and Business Analytics
Human Biology
Human Biology: Physiology and Introductory Anatomy
Observing the Earth from Space
Municipal Engineering

50%

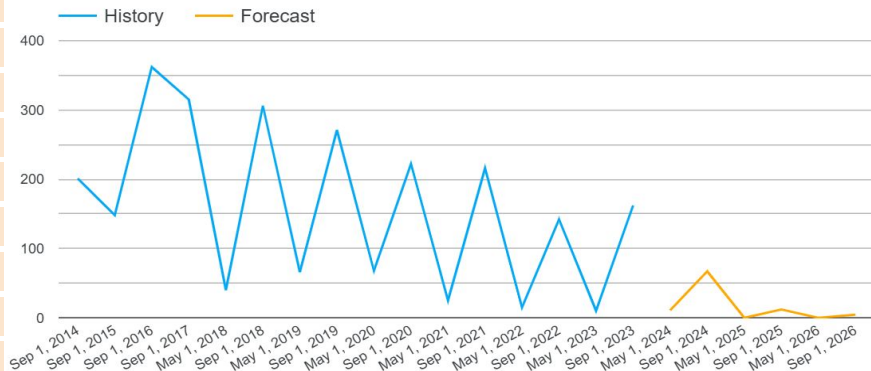
Decrease since 2014

Physical Therapy Foundations
Group Theory Methods in Quantum Mechanics
Feminist Anti-racist Pedagogies
Classical Chinese I (Heritage)
Surface Mining and Design
Machine Components
Entrepreneurial Ehinking for Applied Scientists
Group Inquiry into Educational Practices
Modal Logic
Revolution and Resistance in the Third World

Top 10 Growing Courses with Forecast^[1]



Top 10 Declining Courses with Forecast^[2]



Enrollment Trends

[1] & [2]: Forecast driven by ARIMA PLUS in BigQuery. Seasonality was due to inbalance in UBC's session setting (Winter session covers Sep - Apr while Summer session covers May - Aug).

Correlation Analysis - Method

Step 1: Select Faculty

Focus on **Business faculty** undergraduate courses (100-400 level).

Combine Subject and Course to create a unique course identifier.

Step 2: Top 10 Popular Courses

Rank courses based on **total enrollment (SUM Reported)**.

Select the **top 10 courses**^[1] with the highest student enrollment.

Step 3: Analyze Correlation

Compute **annual average enrollment** for each course.

Standardize enrollment using **Z-score normalization**.

Calculate **Pearson correlation**



When some courses become popular...

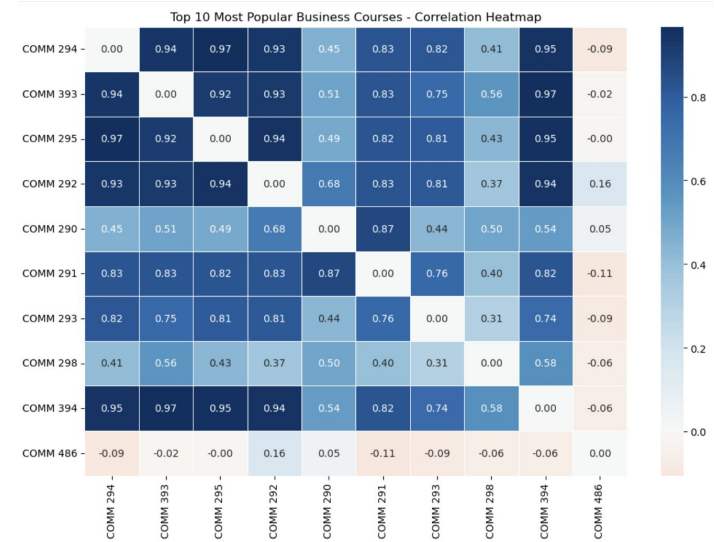
Strong Positive

- If one course becomes more popular, the other is likely to follow.
- UBC should **increase capacity for both courses** rather than just one.
- Consider **co-scheduling or offering course bundles**.

Strong Negative

- Could be due to **scheduling conflicts** or **student specialization choices**.
- Use **"A/B testing"** to check if time conflicts are the cause.
- If not due to scheduling, **overlapping these courses may be efficient**.

Others may change !



Student Performance Analysis-Faculty of Business

4.7 % > 3.9 %



Sauder High Fail Rate Ratio

In Year 2023, 4.7% of all courses in Sauder have a high Fail rate (>10%)

UBC High Fail Rate Ratio



In Year 2023, 3.9% of all courses in UBC have a high Fail Rate (>10%)

Sauder has a Higher High Fail Rate Ratio than University average level!

How can we improve student performance in faculty of business?



Top 10 Fail Rate Courses

Course Title	Fail Rate%
Foundations of Real Estate Mathematics	19.5
British Columbia Real Estate Law and Real Estate	17.81
Foundations of Real Estate Appraisal	16.82
Micro Foundations of Real Estate Economics	16.08
Real Estate Business	14.15
Real Estate Investment Analysis and Advanced Inc.	14
Statistical and Computer Applications in Valuation	11.51
Introductory Management Accounting	11.41
Urban and Real Estate Economics	11.32
Fundamentals of Financial Accounting	10.8

High Fail Rate

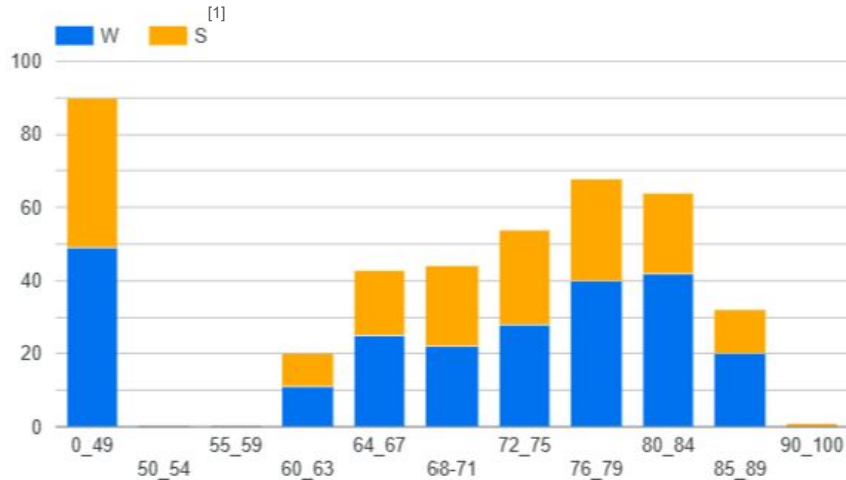


Real Estate



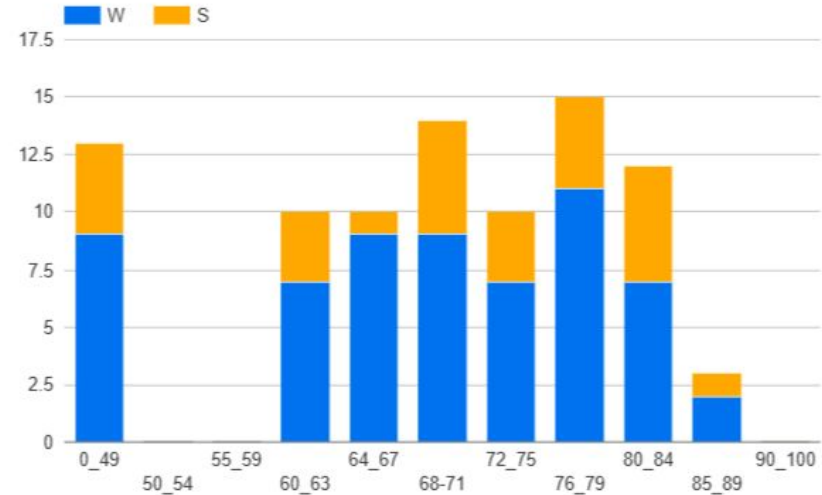
More Details of the Grade Distribution

Foundations of Real Estate Appraisal Grade Distribution Statistical and Computer Applications in Valuation



Fail Rate: 16.8%

High Score Rate: **only 0.24%**



Fail Rate: 15%

High Score Rate: **0%** → **no one gets an A+!**



[1]: W indicates Winter Session (Sep - Apr), S indicates Summer Session (May - Aug).

Course Reviews—What can this indicate?

The course Real Estate Finance in a Canadian Context has a high fail rate: **10%**

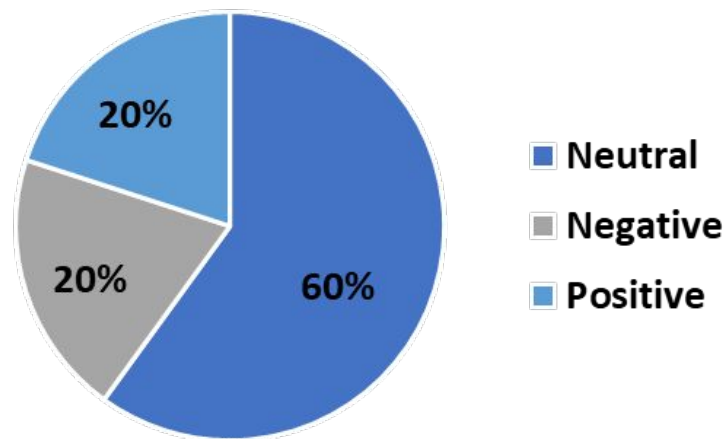
Course Reviews:
Non-positive reviews: 80%!

Courses with high fail rate & non-positive reviews:

What can we do ?

- Extra tutorials
- Additional office hours & TAs
- More academic support

Course Reviews Sentiment^[1]



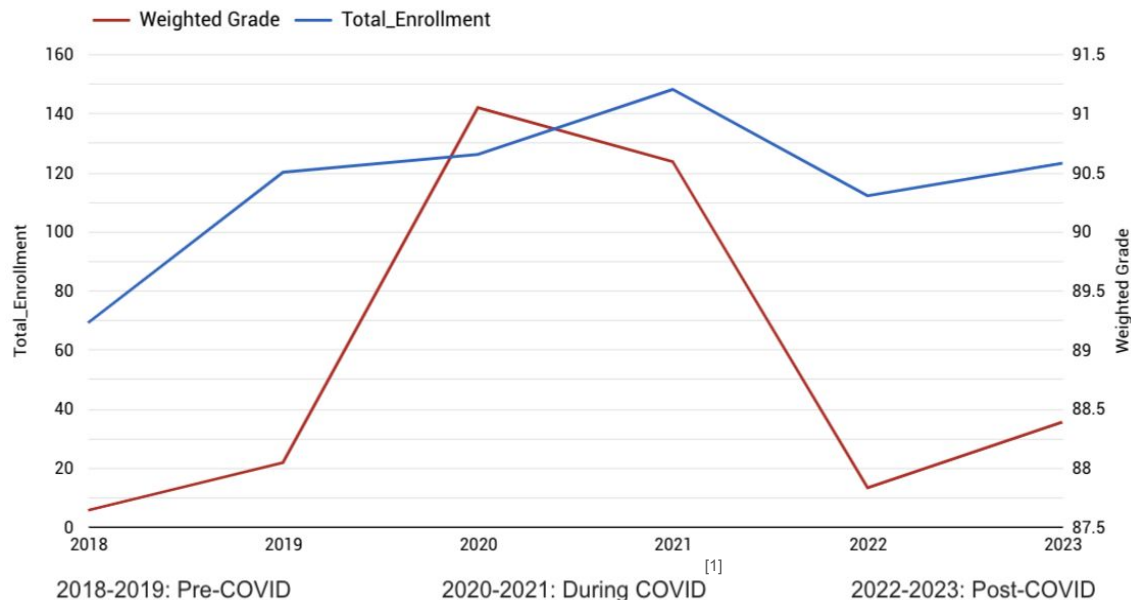
Online Impact on Enrollment & Grade

Weighted Grade and Total_Enrollment Trends

Faculty: Faculty of Science (1) ▾

Full_Title: Data Science (1) ▾

Course_Title: Supervised Learning II (1) ▾



Top 10 Courses - HIGH Enrollment & Performance

- Classes that had **higher enrollments** during online period shows **high student interest** in taking those classes online.
- Classes that had **higher average grade** during online period shows **high student performance** in those classes when online.

Contemporary Social Problems
Ethnography of South Asia
Human Sexual Health
Philosophy After 1800
Introduction to Ecological Anthropology
Questioning Psychological Science in the Media
Analog CMOS Integrated Circuit Design
Introduction to How to Learn a Language
Sex, Gender, and Sexuality in Japanese Literature and Film
Clinical Psychology

178.77% 

Enrollment Growth During Covid (Online)

5.73% 

Average Grade Increase During Covid (Online)



Top 10 Courses - LOW Enrollment & Performance

- Classes that had **lower enrollments** during online period shows **low student interest** in taking those classes.
- Classes that had **lower average grade** during online period shows **low student performance** in those classes.

Structure and Bonding in Chemistry
Directed Studies in Biology
Biomass Fractionation Technology
Critical Reading and Writing in Academic and Professional Registers
Shakespeare
Modern Chinese Fiction in Translation II
Introduction to Language Communities and Variation in Language Practices
Externship Reflection
Assessment of Learning Difficulties
Wood Physics and Mechanics

58.8%



Declined Enrollment During Covid (Online)

4.21%



Declined Average Grade During Covid (Online)



Online Impacts on Enrollment & Performance

High

Potential Explanation:

- Courses rely on digital tools and independent problem-solving
- Lower pressure in online exams
- Recorded lectures can be reviewed

Insights:

- Add online/hybrid options in the future
- Flag potential unmatched difficulty level in course setting when transition to online

Low

Potential Explanation:

- Discussion-heavy, Lab-Based, and Hands-on courses struggled with online mode

Insights:

- Not suitable for online teaching
- Adjust online difficulty if want to offer online class again



Key Take-aways

An intelligent platform empowering course planning & improving

Adjust capacity for High / Low demand courses

Cross-adjust courses sharing same / opposite popularity trends

Add academic support for courses with high fail rate & complaints

Offer hybrid options / Enforce in-person mode for specific courses



Future Work

- **Continuous Data & Model Improvement:** Regularly update datasets and retrain models to enhance accuracy and robustness.
- **Instructor Impact Analysis:** Compare enrollment trends and grade distributions across different professors to identify relevant insights.
- **AI-Driven Insights:** Integrate Gemini AI & Chatbots to provide real-time course highlights and summaries.
- **Expanded User Applications:** Tailor the platform to support diverse users
 - **Professors:** Gain insights for course feedback and future improvement.
 - **Students:** Plan future course enrollments more effectively.
 - **Program Ministers:** Optimize program design based on data-driven insights.



Business Question

How can we use data to assess and improve UBC's course planning?



Business Ambition

Leverage data to shape the future of UBC education



Thank You

Q&A

Hackathon Team 11: Thunder Hackers

Members:

Wendy Liu

Lingxi Li

Heran Ma

Sisly Lyu

Angela Wu



Scan here to explore our dashboard!



Scan here to provide any feedback
for our presentation!



Appendix

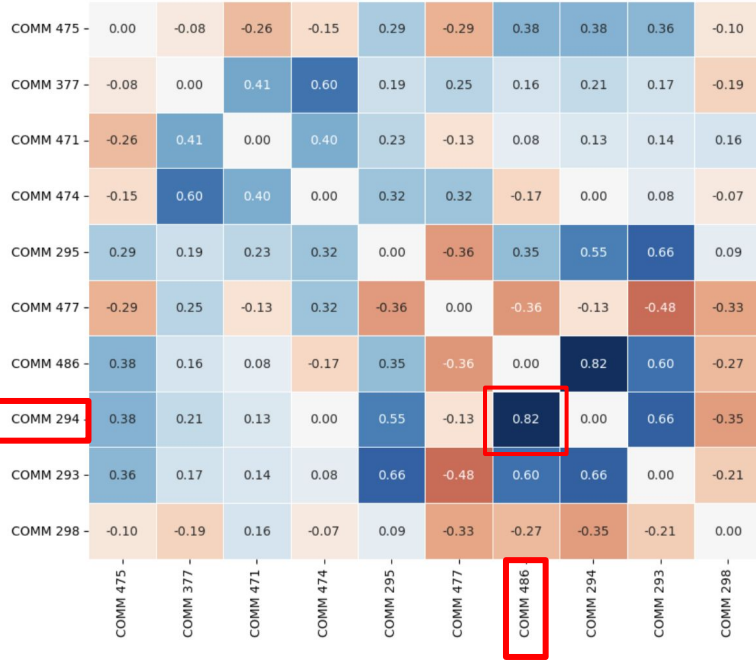
<https://lookerstudio.google.com/reporting/7e2e0181-9af2-43d8-9abe-8bf6526a2803>



COMM 294
COMM 486

Average Grade

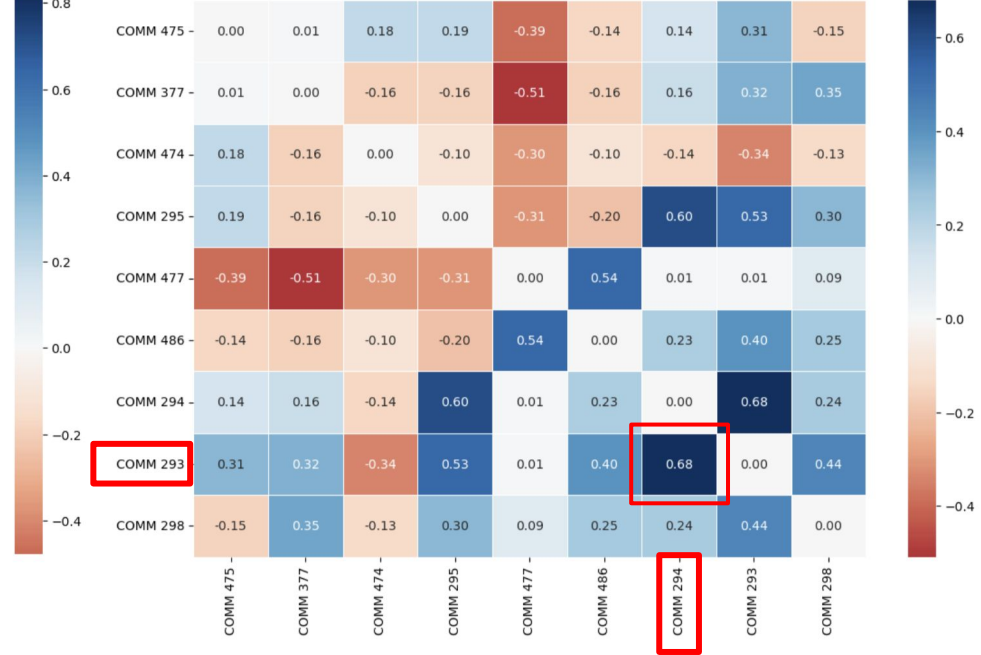
Finance Major Course Average Grade - Correlation Heatmap



Fail Rate

COMM 293
COMM 294

Finance Major Course Fail Rate - Correlation Heatmap



- **Support Needed:** Add tutoring for basic courses to prevent future failures.
- **Fail Risk:** Higher fail rates in one course may lead to issues in related courses.
- **Other Factors:** Consider professor changes and course outline, and other influencing factors.

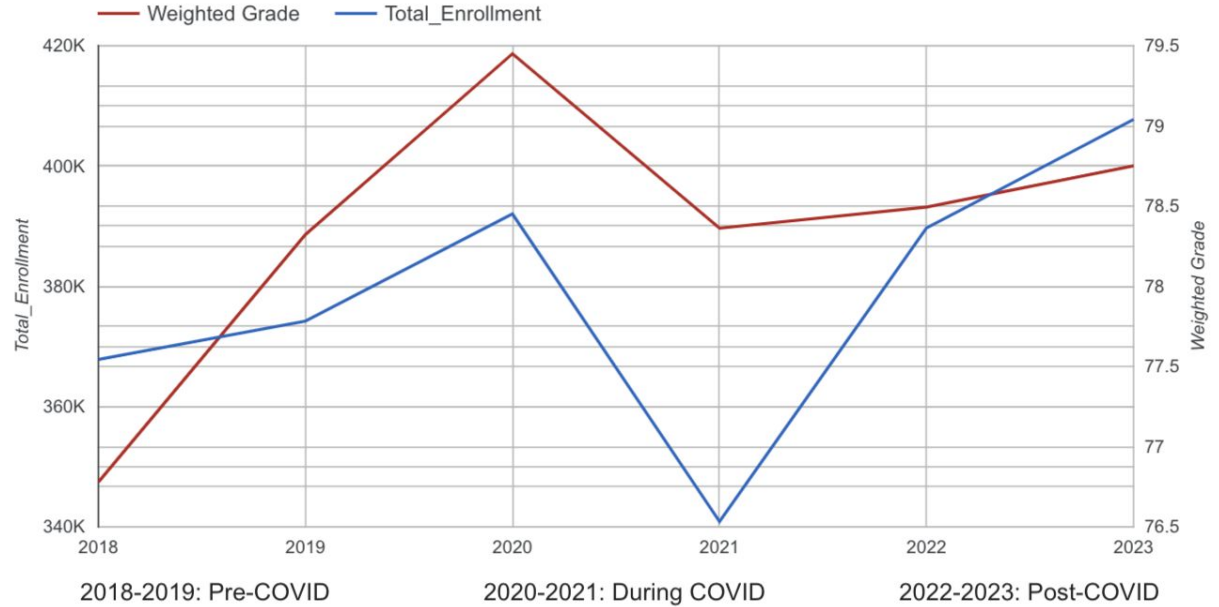


Weighted Grade and Total_Enrollment Trends

Faculty ▼

Full_Title ▼

Course_Title ▼



Weighted Grade and Total_Enrollment Trends

Faculty: Faculty of Commerce and ... (1) ▾

Full_Title: Business Administration:... (1) ▾

Course_Title: Optimal Decision Ma... (1) ▾

